

DSCI 510 – Final Project Progress Report

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Can Qualifying Position Predict Race Results in the 2022–2025 Formula 1 Seasons?

Project Scope Update

The project scope remains consistent with the original proposal. The core research question is unchanged: examining whether qualifying (grid) position can predict race finishing position across the 2022–2025 Formula 1 seasons, which share the same ground-effect aerodynamic regulations. Data from these seasons is now available for analysis, although data cleaning and integration are still required to ensure consistency across sources. The analytical approach — combining exploratory data analysis with a linear regression model — has been confirmed as appropriate given the structure of the data.

Data Sources

OpenF1 API (<https://api.openf1.org/v1>): This REST API provides real-time and historical Formula 1 data in JSON format. Python code has been written to query the /sessions, /positions, and /drivers endpoints. For each Grand Prix, qualifying and race sessions are identified within the same meeting using session_name, and matched based on their shared meeting_key. Position data is fetched for both sessions and merged per driver to create qualifying_position / race_position record pairs. The API has been successfully integrated into the project, and test scripts confirm that data can be retrieved programmatically without authentication.

FastF1 Python Library (<https://docs.fastf1.dev>): This library provides access to official F1 timing data including lap times, tyre compounds, and telemetry. It has been installed and tested. It will be used in later stages for deeper lap-level analysis.

Ergast / Kaggle Dataset: A CSV dataset covering Formula 1 championships from 1950–2025, containing raceId, driverId, constructorId, grid, position, and points fields. This will serve as a validation and supplementary source for historical race results.

Issues / Difficulties

The main difficulty encountered so far is the structure of the OpenF1 positions endpoint: it returns a time-series of position changes throughout a session rather than a single final result per driver. This required additional processing logic to extract the final recorded position for each driver by sorting on the date field and selecting the last value per driver_number. A second challenge is that integrating data across multiple sources requires handling inconsistencies in data availability and structure; the data pipeline is designed to skip incomplete or missing records when necessary. No authentication or rate-limiting issues were encountered with the OpenF1 API.